Week 8 question sheet

Problem sheet

Instructions

Attempt all 11 questions and all 17 scorable parts. The target time is 240 minutes and the total is 100 marks. No calculator is needed: all critical values, square-root approximations and normal probabilities required for the intended hand solution are supplied.

Show the assumptions or approximation conditions used. Distinguish random variables from observed values and parameters or estimands from estimators and estimates. Round reported interval endpoints, probabilities and test statistics to three decimal places unless an exact expression is more informative.

Question 1 — inferential objects and confidence (`S7`, `S8`)

In a population, let $p \in (0, 1)$ be the proportion of students who complete a preparatory task. In a random sample of $n = 120$ students, let X be the number who complete it and define

$$\hat{p} := \frac{X}{120}.$$

The observed count is $x = 84$.

Part A — `W08-Q01-PA` (5 marks; 10 minutes) Identify the population parameter, the estimand, the estimator and the observed estimate. Explain why the estimator $\hat{p}$ is random before sampling but its realised value 84/120 is not random after the data have been observed.

Part B — `W08-Q01-PB` (5 marks; 10 minutes) A 95% confidence-interval method for p is used repeatedly on independent samples of size 120. Assess both statements:

1. "For any one realised interval, there is a 95% probability that the fixed parameter p lies inside it."
2. "In repeated sampling, approximately 95% of the intervals constructed by this method cover p, provided the method's conditions hold."

State the valid frequentist interpretation and explain what is random.

Question 2 — a normal interval with known variance (`S8`)

Part A — `W08-Q02-PA` (6 marks; 15 minutes) Suppose

$$X_1, \dots, X_{36} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 36),$$

where the variance is known. The observed sample mean is $\bar{x} = 74$. Construct and interpret a 95% confidence interval for μ. State why the normal method is exact here. Use $z_{0.975} = 1.96$.

Question 3 — a small-sample Student's t interval (`S8`)

Part A — `W08-Q03-PA` (7 marks; 16 minutes) The observations

$$8, \quad 10, \quad 9, \quad 13, \quad 10$$

are an independent random sample from $\mathcal{N}(\mu, \sigma^2)$, where both μ and $\sigma^2 > 0$ are unknown. Compute $\bar{x}$ and s^2, using

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2,$$

then construct a 95% confidence interval for μ. State the reference distribution, degrees of freedom and conditions. Use $t_{0.975,4} = 2.776$ and $\sqrt{0.7} \approx 0.8367$.

Question 4 — a proportion interval and planned precision (`S8`)

In an independent random sample of 200 eligible applicants, 120 submit a complete form.

Part A — `W08-Q04-PA` (6 marks; 15 minutes) Construct a 90% large-sample confidence interval for the population proportion p. Check the success and failure counts and interpret the interval. Use $z_{0.95} = 1.645$ and $\sqrt{0.6(0.4)/200} \approx 0.0346$.

Part B — `W08-Q04-PB` (5 marks; 15 minutes) For a future independent sample, find the smallest integer n that makes the conservative 95% margin of error no greater than 0.04, using $p(1-p) \leq 1/4$ and $z_{0.975} = 1.96$. Explain why the conservative choice does not require a prior estimate of p.

Question 5 — a one-sided z test (`S9`)

Suppose

$$X_1, \dots, X_{16} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 16),$$

where the variance is known. The observed sample mean is $\bar{x} = 48.5$. A concern is that μ is below 50.

Part A — `W08-Q05-PA` (6 marks; 16 minutes) Formulate H_0 and H_1, calculate the z statistic and its p-value, and make a decision at $\alpha = 0.05$. State the tail convention. Use $\Phi(-1.50) = 0.0668$ and the lower-tail critical value -1.645.

Part B — `W08-Q05-PB` (4 marks; 10 minutes) Interpret the p-value from Part A. Explain why the decision neither proves H_0 nor gives the probability that H_0 is true.

Question 6 — a two-sided t test and its interval (`S8`, `S9`)

Suppose $X_1, \dots, X_{10}$ are independent observations from $\mathcal{N}(\mu, \sigma^2)$, with both parameters unknown. The observed statistics are

$$\bar{x} = 52, \qquad s = 3.$$

Test $H_0 : \mu = 50$ against $H_1 : \mu \neq 50$.

Part A — `W08-Q06-PA` (7 marks; 15 minutes) Calculate the t statistic, state its null distribution and degrees of freedom, calculate the two-sided p-value, and decide at $\alpha = 0.05$. Use $\sqrt{10} \approx 3.162$, $t_{0.975,9} = 2.262$ and $\mathbb{P}(T_9 \geq 2.108) \approx 0.032$.

Part B — `W08-Q06-PB` (5 marks; 12 minutes) Construct the corresponding two-sided 95% confidence interval for μ. Explain why its relation to the null value agrees with the test decision in Part A, and state the conditions needed for this test–interval duality.

Question 7 — a large-sample proportion test (`S9`)

Part A — `W08-Q07-PA` (6 marks; 18 minutes) Let

$$X \sim \text{Binomial}(100, p),$$

where X is the number of successes in 100 independent Bernoulli trials. The observed count is $x = 58$. Test

$$H_0 : p = 0.5 \qquad \text{against} \qquad H_1 : p \neq 0.5$$

at $\alpha = 0.05$, using the null standard error. Check the normal approximation conditions. Use $\Phi(1.60) = 0.9452$ and $z_{0.975} = 1.96$.

Question 8 — Type I error, Type II error and power (`S9`)

Suppose

$$X_1, \ldots, X_{16} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 16),$$

where the variance is known. Consider testing $H_0 : \mu = 50$ against $H_1 : \mu > 50$.

Part A — `W08-Q08-PA` (6 marks; 16 minutes) The rule rejects H_0 when $\bar{X} > 52$. Calculate:

1. the Type I error probability;
2. the Type II error probability $\beta(53)$; and
3. the power at $\mu = 53$.

Use $\Phi(2) = 0.9772$ and $\Phi(-1) = 0.1587$.

Part B — `W08-Q08-PB` (10 marks; 20 minutes) Now choose a sample size n and a rejection rule of the form $\bar{X} > c_n$. Find the smallest integer n for which:

- the significance level is 0.05; and
- the power is at least 0.90 when $\mu = 52$.

Derive both the critical boundary c_n and the sample-size inequality. Use $z_{0.95} = 1.645$, $z_{0.90} = 1.282$ and $\sigma = 4$. Verify that rounding n down fails the power target.

Question 9 — test–interval duality and non-rejection (`S8`, `S9`)

Part A — `W08-Q09-PA` (6 marks; 12 minutes) A two-sided 95% z confidence interval for a population mean, constructed with the same standard error and normal method as a proposed test, is

$$[102.4,\ 107.6].$$

Without the raw data, state the decision at $\alpha = 0.05$ for each test:

$$H_0 : \mu = 100 \quad \text{against} \quad H_1 : \mu \neq 100,$$

and

$$H_0 : \mu = 105 \quad \text{against} \quad H_1 : \mu \neq 105.$$

Explain the duality and correct the claim that non-rejection of the second null means "there is a 95% probability that $\mu = 105$".

Question 10 — earlier Statistics retrieval

Complete both parts without consulting earlier notes.

Part A — `W08-Q10-PA` (`S6`; 4 marks; 10 minutes) Let $X_1, \dots, X_{36}$ be iid random variables with $\mathbb{E}[X_i] = 12$ and $\mathrm{Var}(X_i) = 9$. The population distribution is otherwise unspecified. Find $\mathbb{E}[\bar{X}]$ and $\mathrm{Var}(\bar{X})$. Use the central limit theorem to approximate

$$\mathbb{P}(11.5 \leq \bar{X} \leq 12.5).$$

State why the normal calculation is approximate rather than exact. Use $\mathbb{P}(-1 \leq Z \leq 1) = 0.6826$.

Part B — `W08-Q10-PB` (`S7`; 4 marks; 10 minutes) Let

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p), \qquad 0 < p < 1.$$

For the estimand p, consider the estimator $\hat{p} := \bar{X}$. State its bias and variance, explain why it is consistent, and give the observed estimate when $n = 20$ and $\sum_{i=1}^{20} x_i = 7$.

Question 11 — earlier Mathematics retrieval

Part A — `W08-Q11-PA` (`M10`, `M11`; 8 marks; 20 minutes) For a parameter $a > 0$, solve

$$\max_{x,y \in \mathbb{R}} \{4x - x^2 + 2y\}$$

subject to

$$a - x - y = 0 \qquad \text{and} \qquad y \geq 0.$$

Use the course convention for the inequality $g(x, y) = y \geq 0$:

$$\mathcal{L} = 4x - x^2 + 2y + \mu(a - x - y) + \lambda y, \qquad \lambda \geq 0.$$

State all Kuhn–Tucker conditions. Solve separately for the interior regime $y > 0$ and the corner regime $y = 0$, identify the range of a for each regime, and justify global optimality. Then give the value function $V(a)$ and use the envelope theorem to interpret the equality multiplier $\mu^*(a)$ under the stated encoding $a - x - y = 0$.

Self-test questions

Set aside 24 minutes and attempt all four parts without the reading guide, worked examples, hints or exercise solutions. No calculator is needed: all intended critical values, square-root approximations and normal probabilities are supplied. State the condition, null distribution or repeated-sampling interpretation that supports each conclusion.

The self-test has 20 marks and does not contribute to a checkpoint or mock-examination score.

Questions

`W08-ST01-PA` — **a one-sample t interval (`S8`; 5 marks; 6 minutes)** Suppose $X_1, \dots, X_9$ are independent observations from $\mathcal{N}(\mu, \sigma^2)$, where both parameters are unknown. The observed statistics are $\bar{x} = 15$ and $s = 3$. Construct and interpret a 95% confidence interval for μ. Use $t_{0.975,8} = 2.306$.

`W08-ST02-PA` — **a proportion interval (`S8`; 5 marks; 6 minutes)** In an independent random sample of 100 people, 40 give a specified response. Construct a 95% large-sample confidence interval for the population proportion p. Check the success and failure counts. Use $z_{0.975} = 1.96$ and $\sqrt{0.4(0.6)/100} \approx 0.0490$.

`W08-ST05-PA` — **a one-sided t test and p-value (`S9`; 5 marks; 6 minutes)** Suppose $X_1, \dots, X_{16}$ are independent observations from $\mathcal{N}(\mu, \sigma^2)$, where both parameters are unknown. The observed statistics are $\bar{x} = 21.5$ and $s = 4$. Test $H_0 : \mu = 20$ against $H_1 : \mu > 20$ at $\alpha = 0.05$. Calculate and interpret the p-value. Use $\mathbb{P}(T_{15} \geq 1.5) = 0.077$.

`W08-ST04-PA` — **Type II error and power (`S9`; 5 marks; 6 minutes)** Suppose

$$X_1, \dots, X_4 \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 4),$$

where the variance is known. A test of $H_0 : \mu = 0$ against $H_1 : \mu > 0$ rejects when $\bar{X} > 1.645$. Confirm the Type I error probability, then calculate $\beta(2)$ and the power at $\mu = 2$. Use $z_{0.95} = 1.645$ and $\Phi(-0.355) = 0.3613$.